SHIELDED SIGNAL CABLES

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Updates made since the previous revision
• New cable diameter range added in chapter 10. LIST OF SHIELDED SIGNAL CABLES

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2 / SUPPLIER INFORMATION	D02 - Requirements Document	PDTD - Product

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1. AIM

This document defines the technical requirements of shielded signal cables in SIEMENS GAMESA wind turbines.

2. SCOPE

Siemens Gamesa wind turbines.

3. RELATED DOCUMENTS

CODE	TYPE	TITLE
GD076656	Purchase Technical Specification	Shielded Signal cables

4. DEFINITIONS & ACRONYMS

- T: Ambient Temperature
- STD: Standard Temperature (according to GD076656)
- LT: Low Temperature (according to GD076656)
- HS: High Security cable (according to GD076656)
- HF: Halogen-free (according to GD076656)
- UV: Ultra-violet light or sunlight radiation (according to GD076656)
- TS: Torsion Strength (according to GD076656)

Table 1 - Definitions

SIGNAL CABLE	Type of shielded cable used to transmit signal from different sensors or devices
LT:	Low temperature.
HS:	High Security cable: fulfils with points 11.1, 11.2, 11.3, 11.4, 11.5 and 11.6.
HF:	Halogen free.
STD:	Standard.
Cu:	Conductive copper material.
D:	Outer diameter of cable in mm.
S:	Shield.
Z_0:	Characteristic impedance.
WTG:	Wind Turbine Generator.
N/A:	Not applicable.
Mobile service cable:	Cable on which torsion movement is applied due to nacelle turn. Only tower cables are considered as mobile service.
Fix service cable:	Cable without movement applied.

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Unilay conductor:	Conductor constructed with a central core surrounded by more than one layer of helically laid wires, all layers having a common length and direction of lay.
Rotation cycle:	A full rotation cycle is considered to be completed when the cable is turned from +xº/m to –xº/m and returns to the "0" position.
UV:	Ultra-violet light or sunlight radiation.
Service temperature:	Temperature when the cable is installed and operating.
Storage temperature:	Temperature when the cable is stored and not installed.
G2MW	Gamesa 2.0MW

5. APPLICABLE STANDARS

[1] DIN 47100: Color coding for the identification of cores in cables.

[2] UNE 21089-1: Identificación de los conductores aislados de los cables.

[3] HD 308 S2: Conductor identification for colour coded low voltage multiconductor cables and cords.

[4] VDE 812

[5] IEC 60228: Conductors of insulated cables.

[6] IEC 60811: Common test methods for insulating and sheathing materials of electric cables.

[7] IEC 60332-1-2: Tests on electric and optical fibre cables under fire conditions.

[8] IEC 60332-3: Tests on electric cables under fire conditions: test for vertical flame spread of vertically mounted bunched wires or cables.

[9] IEC 60754: Test on gases evolved during combustion of materials from cables.

[10] IEC 61034: Measurement of smoke density of cables burning under defined conditions.

[11] NES 713: Naval Engineering Standard for the determination of the toxicity index of the products of combustion from small specimens of material.

[12] NFC 20454

[13] UNE-EN 50289-3-16 Communication cables - Specifications for test methods

Other standards could be applicable on request or agreement.

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6. ENVIRONMENT CONDITIONS

6.1. FUNCTIONAL ENVIRONMENT

The signal cables shall transport signals from sensors or actuators to control cabinets.

6.2. PHYSICAL ENVIRONMENT

The cables shall be mainly laid through metal trays and properly fixed with polyamide cable clamps.

In order to allow the nacelle yawing during orientation movement, the cable will be suspended from the nacelle in the upper tower section, so the yawing involves torsion in the cable. Along this tower section, the cable shall bear also its own weight.

Shielded signal cables shall be accessible for maintenance people so they must have a crush resistance equivalent to the service people weight in all temperatures range

Cables must be able to withstand continuous industrial vibrations, abrasion and hits.

Due to the previous conditions, the cables shall be designed to withstand different external agents as ozone, extreme heat cycles, oil, humidity, dust, etc.

6.3. ENVIRONMENT REQUIREMENTS

Table 2 - Environment requirements

Description	Value
Altitude of installation:	Up to 2500m over sea level. (Higher altitudes could be specified on request)
Ambient temperature (STD environment):	-20ºC...+70ºC
Ambient temperature (climate LT environment)	-40ºC...+70ºC
Storage temperature:	-40ºC...+70ºC

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6.4. TECHNICAL REQUIREMENTS

Table 3 - Cables requirements

Description	Value
1. Rated insulation voltage:	Standard: 250/300V Other voltages on request (see special requirements in table 5)
2. Life cycle:	20 years.
3. Conductor:	Flexible electrolytic copper class 5 according to IEC 60228.
4. Insulation features:	
1. Material:	Preferably polyolefin. *Note: Other materials may be approved if they comply with the requirements of this specification and verifications of their suitability for the application are provided.*
2. Colour:	According to DIN 47100 or UNE 21089-1/HD 308 S2. (Other colours could be specified on request)
3. Service temperature:	STD climate: -20ºC...+70ºC LT climate: -40ºC...+70ºC
5. Construction:	Twisted conductors with a twist step normally between 10 and 20 times the overall diameter of the twisted pair. Geometry of the conductors wired shall be in pairs or concentric crowns.
6. Separator:	Polyester band.
7. Drain wire and Shield:	Option 1: Polyester/aluminium band with minimum overlap of 20%. Continuity wire made of unilay flexible cable tin-plated electrolytic copper with a cross-section according to the conductor cross-section. Option 2: Tin-plated copper strand with a minimum cover of 80%.
8. Outer Sheath features:	

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1. Material:	Preferably thermoplastic elastomer or polyolefin. *Note: Other materials may be approved if they comply with the requirements of this specification and verifications of their suitability for the application are provided.*
2. Colour:	Preferably black.
3. Service temperature:	STD climate: -20ºC...+100ºC LT climate: -40ºC...+100ºC
9. Mechanical features:	
1. Torsion strength:	In the upper tower section, the cable must bear the yawing of the nacelle, which involves torsion of 6 turns in a cable length of 20 m. (equivalent to + - 108º/m torsion), bearing its own weight simultaneously in all the defined temperature range. Cycle's number defined in point 9.4.
2. Bending radius:	≤ 12Ð
3. Crush resistance	≥ 1000 N ; plate 10x10 cm; according to UNE_EN 50289-3-5
4. Impact resistance	≥ 10 N/20 impacts, 120mm high; according to UNE-EN 50289-3-6
5. Abrasion resistance	≥ 5000 cycles/1mm steel radio / 4 N. According to UNE-EN 50289-3-7
10. Resistance to external agents:	
1. Oil resistance: CTS GD088841-CTS01D	Cables shall be tested according to IEC 60811-2-1- and IEC 60502-1 Table 22: ○ Standard oil type: IRM 902 ○ Temperature: 100 º C/-2ºC ○ Test duration: 24 hours ○ Maximum variation of tensile strength: +/- 40% ○ Maximum variation of elongation at break: +/- 40%
2. Humidity resistance:	The cable must be resistant to water projection in any direction and to intermittent water immersion according to IEC 60811-1-3 "Water absorption. Clause 9.2. Gravimetric method" and IEC 60502-1 Table 21.

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3. Thermal aging:	Cables shall withstand an aging test according to IEC 60811-1-1 and 60811-1-2. The test conditions shall be 130ºC during 240h (other conditions under agreement). After the test, the maximum variation of tensile strength shall be <20% and the breaking elongation <20%.
4. UV resistance:	A.1. Cables installed outside the wind turbine shall be UV-resistant according to UL1581 point 1200, Xenon equipment. A.2. Indoor installation. In general UV-resistant no required. In case UV-resistant required, UL1581 certificate is no necessary
5. Ozone resistance:	Cables shall be ozone-resistant according to IEC 60811-2-1. chapter 8 and IEC 60502-1 table 17 with the following test conditions: ○ Temperature: 25 ºC ○ Test duration: 24 hours ○ Ozone concentration (by volume): 0,025% to 0,030% No cracks shall appear on the test samples.
11. Fire behaviour:	
1. Flame retardant	According to IEC 60332-1-2.
2. Fire retardant:	According to IEC 60332-3. (only on request)
3. Halogen free:	According to IEC 60754-2. (only on request)
4. Smoke emission:	According to IEC 61034-1 and IEC 61034-2. (only on request)
5. Acid-gas emission:	According to IEC 60754-2. (only on request) pH $\geq$ 4,3 and C$\leq$ 10 μS
6. Toxicity index:	According to NES 713 and NFC 20454. It $\leq$ 4.5 (only on request)

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7. SPECIAL CABLES REQUIREMENTS

Those special requirements included in table 5.

8. DOCUMENTATION

The manufacturer shall send the next information:

- Technical data-sheet (one per different kind of cable; see annex as example)
- List of standards and certification approvals.
- Certifications and results of performed tests.
- Instructions of storage and transport.
- Instructions of installation and maintenance.
- Instructions of security.
- Instructions of environmental recycling.

The Supplier must add a sketch on the datasheet in which the different parts of the cable are shown.

9. ANNEXES

9.1. FEATURES COMBINATION

1. Temperature Requirements:
   - STD: see table 2, point 2
   - LT. see table 2, point 3.

2. Performance requirements:
   - HF (Halogen Free).
   - UV (Ultra-violet resistant).
   - FR (No fire propagation)
   - HS (High security: No Flame propagation + No Fire propagation + Halogen free + Low smoke emission + acid gas emission).
   - MB (Mobile)
   - Performance requirements:

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3. Cables features combination:

Table 4 - List of cables

Number Type	Description	T	HS	HF	UV	TS/MB
1	Standard ambient temperature	STD	NO	NO	NO	NO
2	STD + Halogen free	STD	NO	YES	NO	NO
3	STD + UV-resistant	STD	NO	NO	YES	NO
4	STD + High security	STD	YES	YES	NO	NO
5	STD + Halogen free + UV-resistant	STD	NO	YES	YES	NO
6	STD + High security + UV-resistant	STD	YES	NO	YES	NO
7	STD + Mobile	STD	NO	NO	NO	YES
8	STD + Halogen free + Mobile	STD	NO	YES	NO	YES
9	STD + UV-resistant + Mobile	STD	NO	NO	YES	YES
10	STD + High security + Mobile	STD	YES	YES	NO	YES
11	STD + Halogen free + UV-resistant + Mobile	STD	NO	YES	YES	YES
12	STD + High security + UV-resistant + Mobile	STD	YES	YES	YES	YES
13	Low temperature	LT	NO	NO	NO	NO
14	LT + Halogen free	LT	NO	YES	NO	NO
15	LT + UV-resistant	LT	NO	NO	YES	NO
16	LT + High security	LT	YES	YES	NO	NO
17	LT + Halogen free + UV-resistant	LT	NO	YES	YES	NO
18	LT + High security + UV-resistant	LT	YES	YES	YES	NO
19	LT + Mobile	LT	NO	NO	NO	YES
20	LT + Halogen free + Mobile	LT	NO	YES	NO	YES
21	LT + UV-resistant + Mobile	LT	NO	NO	YES	YES
22	LT + High security + Mobile	LT	YES	YES	NO	YES
23	LT + Halogen free + UV-resistant + Mobile	LT	NO	YES	YES	YES
24	LT + High security + UV-resistant + Mobile	LT	YES	YES	YES	YES
25	Tropical temperature: -0ºC…+70ºC	TT	NO	NO	NO	NO
26	TT + Halogen Free	TT	NO	YES	NO	NO
27	TT + UV-resistant	TT	NO	NO	YES	NO
28	TT + High Security	TT	YES	YES	NO	NO
29	TT + Halogen Free + UV-resistant	TT	NO	YES	YES	NO
30	TT + High Security + UV-resistant	TT	YES	YES	YES	NO
31	TT + Mobile	TT	NO	NO	NO	YES
32	TT + Halogen Free + Mobile	TT	NO	YES	NO	YES
33	TT + UV-resistant + Mobile	TT	NO	NO	YES	YES
34	TT + High Security + Mobile	TT	YES	YES	NO	YES
35	TT + Halogen Free + UV-resistant + Mobile	TT	NO	YES	YES	YES
36	TT + High Security + UV-resistant + Mobile	TT	YES	YES	YES	YES

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10. LIST OF SHIELDED SIGNAL CABLES

Table 5 - Types of cables used

Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
1	2 x 0.75mm^2 + S	-	UNE 21089-1	6 - 7	-
2	3 x 0.34mm^2 + S	-	DIN 47100	6 - 8	-
3	4x0.75mm^2 + S	Blue, Brown, Black, Grey	UNE 21089-1	8 - 10	-
4	4G0.75mm^2 + S	Blue, Brown, Black, Green/Yellow	UNE 21089-1	8 - 10	-
5	4 x 1mm^2 + S	Blue, Brown Black, Grey	UNE 21089-1	8 - 10	-
6	6 x 0.34mm^2 + S	-	DIN 47100	6 - 7.1	-
6.b	6 x 0.34mm^2 + S	-	DIN 47100	6-12	UV
7	10 x 0.34mm^2 + S	-	DIN 47100	8 - 10	-
8	3 x 2 x 0.34mm^2 +S	-	DIN 47100	8 - 10	-
9	8 x 2 x 0.34mm^2 + S		-	10 – 14	Capacitance ≤ 50 pF/m

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
		Blue - Light blue; Red - Light red; Brown - Light brown; Green - Light green; Grey - Light grey; Yellow - Light yellow; White - Black; White - Black			
10	[1 x (2 x 0.5mm^2 + S) + 2 x (2 x 0.34mm^2 + S)] + S	[0.5mm^2]: White, Brown [0.34mm^2]: Green-Yellow; Grey - Pink	DIN 47100	8 - 10	-
11	(8 x 0.75mm^2 + 2 x 1.5mm^2) + S	DIN 47100 (8 x 0.75mm2): 1 White 2 Light Brown 3. Light Green 4. Yellow 5 Grey 6. Pink 7. Black 8. Violet (2 x 1.5 mm2): 1 Light Blue. 2 Red		13 - 14	Rated insulation voltage: 500V

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
12	[2 x (2 x 0.22mm^2 + S)] + S	White, Brown, Green, Yellow	DIN 47100	6 - 8	Z_0 (pair): 100Ω ± 15% (1-20 MHz)
13	[3 x (2 x 0.34mm^2 + S)] + S		DIN 47100	10- 14	-
14	[2 x 0.75mm^2 + (1 x 2 x 0.22mm^2 + S) + 1 x 0.22mm^2] + S	(2x0.75 mm2): White, Brown (2x0.22 mm2): Green, Yellow (0.22 mm2): Grey	DIN 47100	6 - 8	Z_0 (pair): 100 Ω ± 15% (1-20 MHz)
15	4 x (2 x 0.34mm^2 + S)		DIN 47100	10 -14	-
16	[4x(2x0,34mm^2 + S)]+S		UNE 2003	10-12	-
17	[5 x (2 x 0.75mm^2 + S)] + S		DIN 47100	No limit	-
18	6 x 1mm^2 + S		DIN 47100	10 - 12	-
19	10 x 1mm^2 + S		DIN 47100	10 - 14	-
20	15 x 0.75mm +S		DIN 47100	12.5-16	-
21		[0.34 mm^2]: White – Brown; Green – Yellow;		12.5 – 16	Mechanically protected by a steel flexible metal outer mesh that

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
	[3 x (2 x 0.34mm^2 + S) + 7 x 1mm^2] + S Remark: 4 x (2 x 0.34mm^2 + S) + 7 x 1mm^2] + S is also acceptable	Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey, Pink, Blue		(without metallic protection)	ensures C4/H corrosion protection according to standards ISO9223 and ISO14713. Bending radius: 20Ð Z_0 (pair): 120Ω ± 20% (1-100MHz)
22	[3 x (2 x 0.34mm^2 + S) + 11 x 1mm^2] +S Remark: [4 x (2 x 0.34mm^2 + S) + 11 x 1mm^2] +S, is also acceptable	[0.34 mm^2]: White – Brown; Green – Yellow; Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey, Pink, Blue, Red, Black, Violet, grey-pink		12.5 – 16 (without metallic protection)	Mechanically protected by a steel flexible metal outer mesh that ensures C4/H corrosion protection according to standards ISO9223 and ISO14713. Bending radius: 20Ð Z_0 (pair): 120Ω ± 20% (1-100MHz)
23	4 x 0.34 mm2 + S	DIN 47100		6 - 8	-
24	[3 x (2 x 0.5mm^2) + S] +S	DIN 47100		8 - 10.5	-
25	12 x 2 x 0.5mm^2 + S	DIN 47100 BK-RD, BK-WH, BK-GN, BK-BU, BK-YE, BK-BN, BK-OR, RD-YE, RD-BR, RD-WH, RD-GN, RD-BU		13 - 18	-
26	6 x 0.75 mm2 + S	UNE 21089-1		6 - 10	-

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
27	8 x 0.34 mm2 + S		DIN 47100	6 - 8	-
28	8 x 0.75 mm2 + S		DIN 47100	7.5 - 11	-
29	[[(2 x (2 x AWG 24) + S] + [(3 x (2 x 0.22mm^2) + S] + (6 x 1 x 1.5mm^2) + (4 x 1 x 2.5mm^2) + (2 x 1.5mm^2)] + S	[AWG 24]: Orange, Orange – White, Green, Green – White [0.22 mm^2]: White – Brown, Green – Yellow, Grey – Pink [6 x 1.5 mm^2]: Blue, Brown, Black, Grey, Yellow, Green [2.5 mm^2]: Red, White, Orange, Violet [2 x 1.5 mm^2]: Green – Yellow, Green – Yellow		< 22	2 x 2 x AWG 24 : Ethernet 3 x 2 x 0.22 mm2 : Interbus 6 x 1 x 1.5 mm2 (230 V_{AC}) / 300V Voltage service 4 x 1 x 2.5 mm2 (400 V_{AC}) / 450/750 Voltage service 2 x 1.5 mm2 / 500V Voltage service
30	(1x2x0.5 mm^2 +S) +S)	White / Brown	DIN 47100	< 6	Double shield
31	(1x2x0.5 mm^2 +S) +S)	White / Black	DIN 47100	< 6	Double shield
32	2 x 0.22mm^2 + S		UNE 21089-1	4 – 5	-

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
33	2x2xAWG26 + S	Orange/Orange-White/Green/Green-White		6 - 8	-
34	8x26AWG + S	White, Brown, Green, Yellow, Grey, Pink, Blue and Red		6 - 8	UV
35	(3x (2x0.5 mm^2)+S)+S	DIN 47100		6 - 8.7	UV
36	(3GAWG12+(2AWG20+S)) +S	Blue/Brown/Yellow-Green /Orange/Yellow		11.5 - 13	UV
37	4 x 0.5 mm^2	UNE 21089-1 Blue-Brown, Grey-Black		6 - 8	-
38	(4x(2x0,25/AWG24+S)+(2x0.5/AWG21+S))+S	DIN 47100		7.5 - 11	-
39	6 x 0.25 mm^2 + S	DIN 47100		6 - 8	-
40	(2x0,75 mm^2 +2x(2x0,22 mm^2 +S))+S	DIN 47100: (0,75mm2) white –brown; (0,22 mm2) green, yellow, grey		6-8	-
41	4 x (2 x 0.22mm^2 + S)	DIN 47100		8-12	-

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
42	2 x 0.34mm2 + S	DIN 47100		6 - 8	-
43	[3 x (2 x 0.34mm^2 + S) + 5 x 1mm^2] +S	[0.34 mm^2]: White – Brown; Green – Yellow; Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey.		12.5 – 16	Z_0 (pair): 120Ω ± 20% (1-100MHz)
44	[3 x (2 x 0.34mm^2 + S) + 5 x 1mm^2] +S	[0.34 mm^2]: White – Brown; Green – Yellow; Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey.		12.5 – 16 (without metallic protection	Mechanically protected by a steel flexible metal outer mesh that ensures C4/H corrosion protection according to standards ISO9223 and ISO14713. Bending radius: 20Ð Z_0 (pair): 120Ω ± 20% (1-100MHz)
45	12 x 1.5mm^2 + S	Black (wires numbered 1 to 12)		< 20	-
46	12 x 0.75mm^2 + S	DIN 47100		8-12	-
47	48 x 0.34mm^2 + S	Black (wires numbered 1 to 48)		18-25	-
48	8 x 0.22mm^2 + S	1-White 2-Light brown 3- Light green 4-Yellow	DIN 47100	6-7	-

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
		5-Grey 6-Pink 7-Blue 8-Red			
49	[3 x (2 x 0.34mm^2 + S) + 7 x 1mm^2+ 3 x 1.5 mm^2] + S	[0.34 mm^2]: White – Brown; Green – Yellow; Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey, Pink, Blue [3 x 1.5 mm^2]: Brown, Black, Grey		12.5 – 16 (without metallic protection)	Mechanically protected by a steel flexible metal outer mesh that ensures C4/H corrosion protection according to standards ISO9223 and ISO14713. Bending radius: 20Ð Z_0 (pair): 120Ω ± 20% (1-100MHz) 3 x 1.5 mm2 (400 V_{AC}) / 300/500 Voltage service
50	4 x 0.75mm^2 + S	Blue, Brown, Black, Grey	UNE 21089-1	8 - 10	-
51	[3 x (2 x 0.34mm^2 + S) + 7 x 1mm^2] + S Remark: 4 x (2 x 0.34mm^2 + S) + 7 x 1mm^2]+ S is also acceptable	[0.34 mm^2]: White – Brown; Green – Yellow; Grey – Pink; [1 mm^2]: White, Brown, Green, Yellow, Grey, Pink, Blue		12.5 – 16	Z_0 (pair): 120Ω ± 20% (1-100MHz)
52	[2 x (2 x 0.34mm^2 + S)] + S	DIN 47100		7-9	-
53	[3G2.5mm^2 + (2x0.5mm^2+S)]+S	Blue/Brown/Yellow-Green /Orange/Yellow		10-14	Shield option 2. Tin-plated copper strand with a minimum cover of 80%.

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Nº	CABLE TYPE	COLOUR	STANDARD	DIAMETER RANGE [MM]	SPECIAL REQUIREMENTS
54	(3 x 2 x 0.25mm^2) +S	Yellow/green; white/brown; pink/gray		7-9	2 wires twisted together to form a pair, 3 pairs twisted together to form the core **INTERBUS** certified.
55	3 x 0.75mm^2 + S	DIN 47100		7-9	-
56	[2 x 1.5mm^2 + (1 x 2 x 0.22mm^2 + S) + 1 x 0.22mm^2] + S	DIN 47100 (2x1.5 mm2): White, Brown (2x0.22 mm2): Green, Yellow (0.22 mm2): Grey		8 - 10	Z_0 (pair): 100 Ω ± 15% (1-20 MHz)
57	[3G2.5+(2x0.50mm^2+S) +2x0.50mm^2] +S	**3G2.5mm^2**: Azul, marrón, amarillo/verde, amarillo/verde **(2x0.5mm^2) +S**: Naranja, amarillo **2x0.5mm^2**: Rojo, morado		10-18	Cupper comply with UNE 20003. 3G2.5mm2: 0.6/1kV Voltage service 0.5mm2: 500V Voltage service
58	18x0,34mm^2+S	DIN 47100		10-14	-
59	2 x 2,5 mm^2+ S	UNE 21089-1		10-12	-
60	12G1,5 mm^2+ S	Numbered cable wires		13-16	-
61	[2 x (2 x 0.18) + S] + S	DIN47100		6 - 8	Z_0 (pair): 100Ω ± 15% (1-20 MHz)
62	4x1mm2 + S	DIN47100		8 - 10	-
63	4x0,75mm2 + S	DIN47100		8 - 10	-

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11. MAJOR CHANGES COMPARED TO LAST REVISION

Rev	Date	Author	Description
0	04/03/10	JOR/JDELBARRIO /IERVITI	Initial Version.
1	18/01/11	JOR/MTP /IERVITI	Point 3. UNE-EN 50289-3-16 added Point 4.2 PHYSICAL ENVIRONMENT, new redaction used Point 6, 9.1 Tensile strength, requirement written differently Point 6, 9.2 Torsion strength, this test is separated into ambient temperature test and low temperature test Point 6, 9.4 Crush resistance, test method is added Point 6, 9.5 Impact resistance, test method is added Point 6, 9.6 Abrasion resistance, test method is added Point 6, 10.1 Oil resistance, test method is added Point 6, 10.2 Humidity resistance, test method is added Point 6, 10.4 UV resistance, test method is added Point 6, 10.5 Ozone resistance, test method is added Point 6, 11.5 Acid-gas emission, maximum values are added Point 6, 11.6 Toxicity index, maximum value is added Point 9.1 FEATURES COMBINATION is modified. Point 9.2 LIST OF CABLES, the following cables have been created or changed any requirements: 2 x 0.75mm^2 + S 3 x 0.34mm^2 + S 8 x 2 x 0.34mm^2 + S GD076656 r0 -> Inductance: 0.3 - 0.4 mH/km Capacitance: ≥ 40 pF/m GD076656 r1 -> Capacitance ≤ 50 pF/m (8 x 0.75mm^2 + 2 x 1.5mm^2) + S [2 x (2 x 0.22mm^2 + S)] + S [3 x (2 x 0.34mm^2 + S) + 11 x 1mm^2] +S [4 x (2 x 0.34mm^2 + S) + 11 x 1mm^2] 4 x 0.34 mm2 + S (1x2x0.5+S)+S) 2 x 0.22mm^2 + S 2x2xAWG26 + S (3x(2x0.5)+S)+S (3GAWG12+(2AWG20+S))+S Point 9.3 DATASHEET, added Point 9.4 TORSION TEST, test method is added
2	05/10/11	JULJIMENEZ/MTP /IERVITI	Point 9.2 Tensile strength requirement removed

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3	29/01/13	IEUGUI / JULJIMENEZ/MTP /IERVITI	Point 9.2 LIST OF CABLES, the following cables have been created: 12 x 1.5mm^2 + S 12 x 0.75mm^2 S [3 x (2 x 0.34mm^2 + S) + 5 x 1mm^2] +S, with/without mechanical protection (nº 40 and 41).
4	14/05/13	IERVITI / JULJIMENEZ/MTP	Update platforms designation. Point 9.2 LIST OF CABLES, the following cables have been created: ▪ Item 19: G2MW included in WTG model. ▪ Item 38,39,40 and 41 change position ▪ Item 41: (12x0.75mm^2 + S) G2MW and G5MWON are included in WTG model. ▪ Item 42: (48x0.34mm^2 + S) G5MWON are included in WTG model. Item 43: (8x0.22mm^2 + S) Gamesa 850kW is included.
5	25/02/15	IERVITI / JULJIMENEZ/XUBIETO	Point 9.2 LIST OF CABLES, the following cables have been created: Item 44: [3 x (2 x 0.34mm^2 + S) + 7 x 1mm^2+ 3 x 1,5mm^2] + S
6	15/12/15	IERVITI/ JULJIMENEZ/ JLDECARLOS	New cable 4 x 0.75mm^2 + S with new colour
7	20/06/17	BELGONZALEZ/ JULJIMENEZ /FSANTODOMINGO/ JDELBARRIO	Point 9.2 LIST OF CABLES, this cable is created: Item 46: [3 x (2 x 0.34mm2 + S) + 7 x 1mm2] +S
9	08/11/17	JRIPALDA	New cable [2x(2x0.34mm^2+S)]+S
10	06/06/18	XUBIETO	New cable [3G2,5mm² + (2x0,5mm²+S)]+S
11	11/08/18	XUBIETO	Rejected
12	08/01/19	XUBIETO	New cable [3x2x0.25mm^2]+S for interbus New cable 3 x 0.75mm^2 + S for beacon controller. New [2 x 1.5mm^2 + (1 x 2 x 0.22mm^2 + S) + 1 x 0.22mm^2] + S for Sonic LT
13	29/04/2019	SLROTONDO	New cable 4x1mm2 + S added New cable [4x (2x0,34mm^2 + S)] + S added New cable 15x0,75mm^2 + S
14	09/09/2019	SLROTONDO	Included WTG SG2X for cable (3x2x0.25mm^2) + S Included cable 8x26AWG +S (Included for hygrometer DEICING variant SG2X).
15	03/10/2019	CFERMIN	Included cable [3G2.5mm²+(2x0.5mm²)+S+2x0.5mm²]+S (for SG4.X Orga Beacon with Controller Configuration) Modificated the standard of cable [4x(2x0.34mm^2 + S)]+S (nº 15), and [15 x 0.75mm^2]+S (nº 19). New standard DIN 47100
16	20/05/2020	IIRIGOYEN	New cable 18x0.34mm2+ S added

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17	02/07/2020	SLROTONDO/ MABLANCO	Section 9.1 Features combination - Type cables chart format updated Section 9.2 List of cables. - - Cable Nº3 last colour updated from Green/Yellow to Grey - Cable Nº4 created:4G0.75mm2 - Added (1x2x0.5mm2 +S) +S) of wires colors white/ black according to number 25 of DIN 47100. - List numbers updated from the nº 4 to the end - WTG MODEL column deleted.
18	23/07/2020	IIRIGOYEN	Section 9.2 List of cables. - Cable Nº22 color updated from orange to grey-pink according to DIN47100 - Cable Nº59 2x2,5 mm2 + S created Cable Nº60 12G1,5 mm2 + S created
19	02/06/2021	MABLANCO	• New format • Section 9.1 corrected cables “HS=YES + HF=NO” to “HS=YES + HF=YES”
20	11/11/2021	IIRIGOYEN	SC included in following chapters according to “PRO-46240 Identification and Documentation of Special Characteristics” Chapter 6.4
21	04/03/2022	IIRIGOYEN	New cable types added in chapter 10. LIST OF SHIELDED SIGNAL CABLES • Included cable type [2 x (2 x 0.18) + S] + S (DIN 47100) • Included cable type 4x1mm2 + S (DIN47100) • Included cable type 4x0,75mm2 + S (DIN47100)
22	28/08/2023	XMARTINEZ	New cable diameter range added in chapter 10. LIST OF SHIELDED SIGNAL CABLES • Cable num. 36: (3GAWG12+(2AWG20+S))+S 11.5-13mm diameter range.

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